

# Full Molecular Hamiltonian

From Classical to Quantum: the exact non-relativistic quantization

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# Roadmap: four stages

We build the **exact** non-relativistic molecular Hamiltonian in four stages:

1. **Classical** — lab-frame  $H$ ; separate the centre of mass.
2. **Body-Fixed** — co-rotating frame + Lagrangian  $\Rightarrow$  classical molecular  $H$ .
3. **Quantize** — Podolsky rule, anomalous algebra, Watson pseudopotential.
4. **Quantum**  $\hat{H}$  — the exact non-relativistic molecular Hamiltonian.

# Outline

Motivation

Classical Cartesian Hamiltonian

Nuclear centre-of-mass separation

Body-fixed frame and the Lagrangian

The quantization problem

Podolsky quantization

Anomalous body-fixed angular momentum

Watson pseudopotential

The complete quantum Hamiltonian

Summary

# Motivation

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# Why is the molecular Hamiltonian non-trivial?

The lab Hamiltonian couples *all* particles through the Coulomb potential:

$$H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

To do molecular physics we want to separate

$$\underbrace{\text{translation}}_{\text{COM}} \mid \underbrace{\text{rotation}}_{\text{Euler angles } \Omega} \mid \underbrace{\text{vibration}}_{\text{normal coords } Q} \mid \underbrace{\text{electrons}}_{\bar{\mathbf{r}}_i}.$$

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## The catch

Rotation + vibration are *curvilinear* coordinates. Naively quantizing  $p \rightarrow -i\hbar \partial$  is **not** Hermitian there. Getting it right forces the Podolsky construction, an *anomalous* angular-momentum algebra, and the Watson pseudopotential.

# What we will not approximate

We keep everything exact until the very end:

- **No** Born–Oppenheimer / clamped nuclei.
- **No** neglect of *mass polarization*.
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The *only* ingredients are:

1. non-relativistic dynamics;
2. the Eckart frame (a *convention*, not an approximation);
3. parametrizing the nuclear shape by  $3N_{\text{nuc1}} - 6$  normal coordinates (exact).

*The endpoint is the exact quantum  $\hat{H}_{\text{rve}}$  — the rigorous starting point for any later approximation.*



# Classical Cartesian Hamiltonian

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## The starting point (lab frame)

$N_{\text{nucl}}$  nuclei  $(M_j, Z_j e, \mathbf{R}_j, \mathbf{P}_j)$  and  $N_{\text{elec}}$  electrons  $(m_e, -e, \mathbf{r}_i, \mathbf{p}_i)$ , all in the space-fixed frame:

$$H = \underbrace{\sum_j \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_i \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \quad (1.1)$$

$$V = \frac{1}{4\pi\epsilon_0} \left( \frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

Canonical brackets  $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk} \delta_{\alpha\beta}$ ,  $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij} \delta_{\alpha\beta}$ .

Everything that follows is an exact rewriting of (1.1)–(1.2).

## Nuclear centre-of-mass separation

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## New coordinates: nuclear COM + relative

Split off the nuclear centre of mass and refer electrons to it:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \quad M_N = \sum_j M_j, \quad (2.1)$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (2.2)$$

The relative nuclear vectors are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \quad (2.3)$$

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A point ( $F_2$ -)transformation gives the conjugate momenta [dual constraint  $\sum_k \mathbf{P}'_k = 0$ ]:

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N} (\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) + \mathbf{P}'_j, \quad \mathbf{p}_i = \mathbf{p}'_i} \quad (2.7)$$

with  $\mathbf{P}_{\text{cm}} = \sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i$  the *total* momentum.

## Hamiltonian after the transformation

Insert (2.7) and expand the nuclear kinetic energy, keeping  $\mathbf{P}_{\text{cm}}$  *arbitrary*:

$$H = \underbrace{\frac{\mathbf{P}_{\text{cm}}^2}{2M_N}}_{(A) \text{ wrong mass}} - \underbrace{\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}_i}{M_N}}_{(B) \text{ cross-coupling}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass polarization}} + \sum_j \frac{\mathbf{p}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V. \quad (2.10)$$

### Why the mass-polarization term?

It is the *price* of referring electrons to the *nuclear* COM (not the total COM): the electron momenta no longer combine additively with  $M_N$ . We keep it exactly — it is never dropped.

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Artifacts (A) and (B) appear only because the split used the *nuclear* COM. We remove both **exactly, for any  $\mathbf{P}_{\text{cm}}$**  — with *no* need to set  $\mathbf{P}_{\text{cm}} = 0$ .

## Eliminating the cross-coupling, for any $\mathbf{P}_{\text{cm}}$

A single classical canonical transformation generated by  $G = \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{r}'_i$  removes *both* artifacts at once:

$$\tilde{\mathbf{r}}'_i = \mathbf{r}'_i, \quad \tilde{\mathbf{p}}'_i = \mathbf{p}'_i - \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}}, \quad \tilde{\mathbf{R}}_{\text{cm}} = \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}'_i. \quad (2.15)$$

### Why this works

It relabels the COM from *nuclear* to *total* — a Galilean boost. The wrong mass (A) is corrected  $M_N \rightarrow M_{\text{tot}}$  and the cross-coupling (B) is killed, *exactly and for any*  $\mathbf{P}_{\text{cm}}$ . So we never assume  $\mathbf{P}_{\text{cm}} = 0$ .



# The molecular Hamiltonian $\tilde{H}$ : correct mass, no cross-coupling

Exact for *any*  $\mathbf{P}_{\text{cm}}$ ,  $\tilde{H}$  splits into a free COM term and the **rotation–vibration–electronic** Hamiltonian  $H_{\text{rve}}$ :

$$\tilde{H} = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + H_{\text{rve}}, \quad H_{\text{rve}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\tilde{\mathbf{p}}_i'^2}{2m_e} + \frac{1}{2M_N} \left( \sum_i \tilde{\mathbf{p}}_i' \right)^2 + V. \quad (2.22)$$

- COM term carries the *correct* total mass  $M_{\text{tot}} = M_N + N_{\text{elec}}m_e$ ;  $\mathbf{P}_{\text{cm}}$  is a decoupled, conserved spectator.
- $H_{\text{rve}}$  = nuclear KE + electron KE + mass polarization + Coulomb.

$H_{\text{rve}}$  is what we carry into the body-fixed frame.

## Body-fixed frame and the Lagrangian

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## Body-fixed frame: rotate with the nuclei

Introduce a rotation  $\mathbf{S}(\Omega)$  (Euler angles  $\Omega = \phi, \theta, \chi$ ) and body-fixed (BF) coordinates:

$$\mathbf{R}'_j = \mathbf{S}(\Omega) \bar{\mathbf{R}}_j(Q), \quad \mathbf{r}'_i = \mathbf{S}(\Omega) \bar{\mathbf{r}}_i. \quad (3.1)$$

The *Eckart* conditions fix  $\mathbf{S}$  from the *nuclei*; the nuclear shape is rectilinear mass-weighted **normal coordinates**  $Q_b$ :

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb} Q_b, \quad \sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}. \quad (3.1a)$$

The BF angular velocity  $\omega$  is defined by

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma} \omega_\gamma. \quad (3.2)$$

## A Lagrangian that keeps electrons in momentum form

Start from  $\tilde{H}$  (2.22). We need an action principle for the body-fixed *nuclear* coordinates — but the electron kinetic energy + mass polarization  $H_e = \sum_i \tilde{\mathbf{p}}_i'^2 / 2m_e + \frac{1}{2M_N} (\sum_i \tilde{\mathbf{p}}_i')^2$  is a *rotational scalar*. So we Legendre-transform only the COM and nuclear momenta and **keep the electrons in momentum form**, giving the Lagrangian

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\tilde{\mathbf{R}}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j'|^2 - H_e(\tilde{\mathbf{p}}_i') - V. \quad (3.6)$$

### Why keep electrons in momentum form?

It avoids converting the electron kinetic energy to velocity form and back: the electrons ride along as the scalar  $H_e$ . So  $\mathcal{L}$  is a Lagrangian in the nuclear/COM coordinates  $(\tilde{\mathbf{R}}_{\text{cm}}, \mathbf{R}_j')$  and carries the electrons through their momenta  $(\mathbf{r}_i', \tilde{\mathbf{p}}_i')$ .

## Body-fixed Lagrangian

$H_e$  is built from  $\tilde{\mathbf{p}}'_j \cdot \tilde{\mathbf{p}}'_j$ ; under the passive rotation  $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T \tilde{\mathbf{p}}'_i$  it is *unchanged* in form. Substituting the BF velocities  $\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j)$ ,

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\bar{\mathbf{R}}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \left( \sum_i \bar{\mathbf{p}}_i \right)^2 - V. \quad (3.7)$$

Only the *nuclear* kinetic term carries  $\boldsymbol{\omega}$ ;  $V$  is now manifestly  $\mathbf{S}$ -independent.

## Expand the nuclear term

The vector identity (for any  $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega}$ )

$$\frac{m}{2} |\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2 = \underbrace{\frac{m}{2} \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis}} + \underbrace{\frac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}} \quad (3.9c)$$

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applied to the nuclei ( $m \rightarrow M_j$ ,  $\mathbf{a} \rightarrow \bar{\mathbf{R}}_j$ ) gives the assembled Lagrangian

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \underbrace{\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N}_{\text{nuclear}} - H_e - V. \quad (3.13)$$

The rotational kernel is the *nuclear* inertia  $\mathbf{I}_N$  alone — no electronic inertia, because the electrons were never expanded in  $\boldsymbol{\omega}$ .

# Canonical momenta and the total angular momentum

**Nuclear** angular momentum (from  $\partial\mathcal{L}/\partial\omega$ ):

$$\mathbf{J}_N = \mathbf{I}_N\omega + \mathbf{L}_N^{\text{vib}}. \quad (3.15)$$

The BF frame rotates nuclei *and* electrons, so the **total** angular momentum is the sum [ $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ ]:

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}} = \mathbf{I}_N\omega + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \quad (3.16)$$

Vibrational momentum, with the **Coriolis vectors**  $\sigma_b$ :

$$P_b = \omega \cdot \sigma_b + \dot{Q}_b, \quad \sigma_b = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}. \quad (3.17\text{a,b})$$



## The key identity

Subtract the electronic part from (3.16):

$$\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q) \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}. \quad (3.19)$$

### Why this matters

The right side is *purely nuclear*: the rotational kernel is  $\mathbf{I}_N$ , **not** the full inertia. All electronic + mass-polarization content sits on the left, in the kinematic shift  $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}}$ .

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*Cross-check*: the velocity-form route instead generates the electronic inertia  $\mathbf{I}_e, \mathbf{I}_X$  in *both*  $\mathbf{J}$  and  $\mathbf{L}_{\text{elec}}$  — they cancel exactly, reproducing (3.19). The Lagrangian makes the cancellation automatic.

## Watson reciprocal inertia

Invert (3.19) for  $\omega$ : write  $\mathbf{L}_N^{\text{vib}} = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^T\omega$  with

$$\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b \quad (\text{field-free vibrational ang. mom.}), \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^T = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T. \quad (3.20a)$$

Then  $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\omega + \boldsymbol{\pi}$ , so

$$\boxed{\omega = \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}), \quad \boldsymbol{\mu} \equiv (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}.} \quad (3.20)$$

$\boldsymbol{\mu}$  is the **Watson reciprocal-inertia tensor** — built from *nuclear* quantities only, with no Wilson *G*-matrix.

# The exact classical molecular Hamiltonian

Legendre back; the rovibrational kinetic energy becomes

$$\frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2:$$

$$\begin{aligned} H_{\text{rve}} = & \frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(Q)(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2 \\ & + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left( \sum_i \bar{\mathbf{p}}_i \right)^2 + V_{NN} + V_{Ne} + V_{ee}. \end{aligned} \quad (3.22)$$

Exact, classical, in BF + Eckart coordinates. **Now we quantize.**

# The quantization problem

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## Why naive quantization fails in curvilinear coordinates

Label the configuration coordinates and write the classical kinetic energy as a quadratic form:

$$\{q^A\} = \{\phi, \theta, \chi, Q_1, \dots, Q_{3N_{\text{nuc}}-6}, \bar{\mathbf{r}}_1, \dots\}, \quad T = \frac{1}{2} g^{AB}(q) \pi_A \pi_B. \quad (4.1, 4.2)$$

The metric  $g_{AB}$  inherits the block structure of (3.22): Watson  $\mu(Q)$  (rotation), identity (vibration), Cartesian (electrons).

### The problem

The Euler angles and  $Q$  are *curvilinear*. With the curvilinear volume element  $\sqrt{|g|} d^n q$ , the naive quantization  $\hat{\pi}_A = -i\hbar\partial_A$  makes  $\frac{1}{2}g^{AB}\hat{\pi}_A\hat{\pi}_B$  **non-Hermitian** ( $g^{AB}$  depends on  $q$  and does not commute with  $\partial_A$ ).

## Podolsky quantization

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## Podolsky quantization: the recipe

Three steps from the classical  $T = \frac{1}{2}g^{AB}\pi_A\pi_B$  to the exact quantum  $\hat{T}$ :

1. **Write the unique Hermitian operator** consistent with the curvilinear volume element  $\sqrt{|g|} d^n q$  — the Podolsky / Laplace–Beltrami form.
2. **Identify**  $\sqrt{|g|}$  for the molecule from the block structure of  $g_{AB}$ .
3. **Flatten the vibrational measure** via  $\psi \rightarrow \psi_W$  — the  $Q$ -Jacobian generates the Watson pseudopotential.

The anomalous BF angular-momentum algebra ensures the rotational sector contributes *no* extra potential.



# The Podolsky / Laplace–Beltrami prescription

**Step 1.** The unique kinetic operator self-adjoint with respect to the volume element  $\sqrt{|g|} d^n q$  is the Laplace–Beltrami operator:

$$\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{|g|}} \partial_A (\sqrt{|g|} g^{AB} \partial_B). \quad (4.3)$$

Equivalently, with the *Hermitian* momenta  $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ , one has  $\hat{T} = \frac{1}{2} \hat{\pi}_A g^{AB} \hat{\pi}_B$ .

## Why this form?

The  $\sqrt{|g|}$  factors symmetrize the operator about the curved measure. The price: a residual  $Q$ -dependent  $c$ -number — the **Watson pseudopotential** — once we flatten the measure.

## The molecular volume element

**Step 2.** Building  $g_{AB}$  from the velocity-form kinetic energy and reducing the determinant by block-matrix identities gives

$$\boxed{\sqrt{|g|} d^n q = \underbrace{\sin \theta d\phi d\theta d\chi}_{\text{Euler-angle measure}} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)} \prod_b dQ_b}_{\text{rot.-vib. Jacobian}} \cdot \underbrace{\prod_i d^3 \bar{\mathbf{r}}_i}_{\text{flat}}} \quad (4.4)$$

The Jacobian carries the **Watson modified inertia**  $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ , *not* the bare  $\mathbf{I}_N$ : the rotation–vibration Coriolis coupling reduces  $\det \mathbf{I}_N \rightarrow \det \mathbf{I}'$ .

## Rescaling to a flat measure

**Step 3.** Rescale the wavefunction by the Q-Jacobian  $\gamma \equiv \det \mathbf{I}'$  to work on the flat measure  $d\mu_W = \sin \theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3 \mathbf{r}_i$ :

$$\psi_W \equiv (\det \mathbf{I}'(Q))^{1/4} \psi, \quad \hat{T}_W = \mathcal{J}^{1/2} \hat{T} \mathcal{J}^{-1/2}, \quad \mathcal{J} = (\det \mathbf{I}')^{1/2}. \quad (4.5)$$

### Why rescale?

$\psi_W$  is normalised with the *flat* vibrational measure, so  $\hat{P}_b = -i\hbar \partial_{Q_b}$  is Hermitian. The similarity does not commute through  $\gamma(Q)$  — the leftover is exactly the pseudopotential  $\hat{U}(Q)$ .

## **Anomalous body-fixed angular momentum**

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## Body-fixed angular momentum has a sign flip

Total  $\mathbf{J}$  is an SF observable; its BF components are projections through  $\mathbf{S}(\Omega)$ :

$$\hat{J}_{\alpha}^{\text{BF}} = \mathbf{S}_{\beta\alpha} \hat{J}_{\beta}^{\text{SF}}. \quad (4.6)$$

But  $\mathbf{S}(\Omega)$  depends on the Euler angles that  $\hat{J}^{\text{SF}}$  differentiates. Commuting  $\hat{J}^{\text{SF}}$  past  $\mathbf{S}$  adds terms that *outweigh and flip* the usual algebra:

$$\boxed{[\hat{J}_{\alpha}^{\text{BF}}, \hat{J}_{\beta}^{\text{BF}}] = -i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_{\gamma}^{\text{BF}}.} \quad (4.7)$$

The famous **anomalous commutator** (Klein; Van Vleck): note the  $-$  sign.

## Normal vs. anomalous; commuting sectors

The electronic angular momentum is built intrinsically from BF electron operators — *normal* algebra:

$$[\hat{L}_{\text{elec},\alpha}, \hat{L}_{\text{elec},\beta}] = +i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_{\text{elec},\gamma}, \quad [\hat{J}_{\alpha}^{\text{BF}}, \hat{L}_{\text{elec},\beta}] = 0. \quad (4.8, 4.9)$$

### Why it (barely) matters

The sign flip controls the ordering of  $\hat{J}_{\alpha}\hat{J}_{\beta}$  in the kinetic operator. But contracted with the *symmetric*  $\mu_{\alpha\beta}$  the antisymmetric commutator drops out ( $\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$ ) — so it contributes *no* extra potential. We use this next.

## Watson pseudopotential

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## The general curvilinear extra-potential

Conjugating the Laplace–Beltrami operator  $\hat{T}_{LB} = -\frac{\hbar^2}{2} g^{-1/2} \partial_A (g^{1/2} g^{AB} \partial_B)$  by  $g^{1/4}$  and collecting terms (first-order pieces cancel by symmetry of  $g^{AB}$ ) gives the exact, coordinate-general result

$$\hat{T}_W = \frac{1}{2} \hat{p}_A g^{AB} \hat{p}_B + U, \quad U = \frac{\hbar^2}{8} \left[ \frac{1}{4} g^{AB} (\partial_A \ln g) (\partial_B \ln g) + \partial_A (g^{AB} \partial_B \ln g) \right]. \quad (4.10d)$$

### Why $\hbar^2/8$ ?

The two half-powers  $g^{\pm 1/4}$  of the rescaling each contribute  $\frac{1}{2} \partial \ln g$ ; their product/derivative is  $\mathcal{O}(\hbar^2)$  with prefactor  $\frac{1}{8}$ .



# The molecular metric and its determinant

In the body-frame momentum basis  $(\hat{J}_\alpha, \hat{P}_b)$  [with  $\pi = \sigma^\top \hat{P}$ ], read off

$$g^{AB} = \begin{pmatrix} \mu & -\mu\sigma^\top \\ -\sigma\mu & \mathbb{1} + \sigma\mu\sigma^\top \end{pmatrix}, \quad \det g^{AB} = \det \mu \quad (4.10\text{e,f})$$

(block-matrix determinant identity). Hence  $g = 1/\det \mu = \det \mathbf{I}' \equiv \gamma(Q)$ . Since the  $\hat{J}_\alpha$  already carry  $\sin \theta$ , only  $\gamma(Q)$  is rescaled: set  $\ln g \rightarrow \ln \gamma$  in (4.10d).

## Rotational sector: no $c$ -number

The only ordering ambiguity is in  $\hat{J}_\alpha \hat{J}_\beta$ . Split and use the anomalous algebra (4.7):

$$\frac{1}{2} \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta = \frac{1}{4} \mu_{\alpha\beta} \{ \hat{J}_\alpha, \hat{J}_\beta \} - \frac{i\hbar}{4} \mu_{\alpha\beta} \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma. \quad (4.10g)$$

Symmetric  $\mu_{\alpha\beta} \times$  antisymmetric  $\epsilon_{\alpha\beta\gamma} = 0 \Rightarrow$  the commutator term vanishes. The rotational kernel is Hermitian as written, with no extra potential.

## Vibrational sector: the rescaling, step by step

With  $g^{ab} = \delta_{ab}$  but measure  $\gamma(Q)$ , the Podolsky vibrational operator is

$\hat{T}_{\text{vib}} = -\frac{\hbar^2}{2}\gamma^{-1/2}\sum_b\partial_{Q_b}(\gamma^{1/2}\partial_{Q_b})$ . Conjugate by  $\gamma^{1/4}$ , working outward (one product rule per line):

$$\begin{aligned}\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{-1/4}[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W], \\ \gamma^{1/2}\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{1/4}[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W], \\ \partial_{Q_b}(\gamma^{1/4}[\dots]) &= \gamma^{1/4}[\partial_{Q_b}^2\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\psi_W - \tfrac{1}{4}(\partial_{Q_b}^2\ln\gamma)\psi_W].\end{aligned}$$

The cross terms  $\pm\frac{1}{4}(\partial\ln\gamma)\partial\psi_W$  cancel. Result:

$$\gamma^{1/4}\hat{T}_{\text{vib}}\gamma^{-1/4} = \tfrac{1}{2}\sum_b\hat{P}_b^2 + U_{\text{vib}}, \quad U_{\text{vib}} = \frac{\hbar^2}{8}\sum_b\left[\partial_{Q_b}^2\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\right]. \quad (4.10k)$$

## Collapse to the Watson trace

Two exact identities finish the job. **Jacobi:**

$$\partial_b \ln \gamma = \text{tr}(\boldsymbol{\mu} \partial_b \mathbf{l}'), \quad \partial_b \boldsymbol{\mu} = -\boldsymbol{\mu}(\partial_b \mathbf{l}')\boldsymbol{\mu}. \quad (4.10\text{I})$$

**Linearity:**  $\mathbf{l}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\text{T}$  with  $\boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a$ , so  $\partial_b \mathbf{l}'$  is built from *constant*  $\zeta$ 's. Substituting, the gradient terms cancel and the Coriolis sum rules collapse everything to a single trace:

$$\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^3 \mu_{\alpha\alpha}(Q). \quad (4.11)$$

A pure *c*-number from the *measure* — the electrons do not modify it.

# The complete quantum Hamiltonian

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## Electronic kinetic operator

In BF Cartesian electron coordinates  $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\mathbf{r}_i}$  is already Hermitian on the flat measure, so the electron + mass-polarization piece quantizes directly:

$$\hat{T}_e = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left( \sum_i \hat{\mathbf{p}}_i \right)^2 = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2\mu_e} + \frac{1}{M_N} \sum_{i<l} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_l. \quad (4.12, 4.13)$$

The diagonal part renormalizes the electron mass,  $\frac{1}{m_e} + \frac{1}{M_N} = \frac{1}{\mu_e}$ ; the off-diagonal two-electron term is the mass polarization *sensu stricto* (nuclear recoil).

# The exact non-relativistic molecular Hamiltonian

Collecting the rovibrational kernel (3.22), the Watson pseudopotential (4.11), and  $\hat{T}_e$ :

$$\begin{aligned}
 \hat{H}_{\text{rve}} = & \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) && \text{rotation-vibration-electron} \\
 & + \frac{1}{2} \sum_b \hat{P}_b^2 && \text{vibration} \\
 & + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left( \sum_i \hat{\mathbf{p}}_i \right)^2 && \text{electrons + mass pol.} \\
 & - \frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}(Q) && \text{Watson pseudopotential} \\
 & + V_{NN} + V_{Ne} + V_{ee} && \text{Coulomb}
 \end{aligned} \tag{4.14}$$

$$d\mu_W = \sin\theta \, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\mathbf{r}_i.$$

# Summary

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## The journey, in four stages

1. **Classical**: lab  $H$  (1.1)  $\rightarrow$  frame-independent  $\tilde{H}$  (2.22), COM split off exactly for any  $\mathbf{P}_{\text{cm}}$ .
2. **Body-Fixed**: Lagrangian (electrons stay canonical)  $\rightarrow$  key identity  $\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}$   
 $\rightarrow$  classical  $H_{\text{rve}}$  (3.22) with Watson  $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}$ .
3. **Quantize**: Podolsky operator (4.3) + rescaling (4.5); anomalous algebra  
 $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ ; Watson pseudopotential  $\hat{U} = -\frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}$ .
4. **Quantum  $\hat{H}$** : the exact non-relativistic  $\hat{H}_{\text{rve}}$  (4.14).

## What we did — and did not — assume

$\hat{H}_{\text{rve}}$  (4.14) is **exact**. The only inputs were:

- non-relativistic dynamics;
- the Eckart frame (a convention);
- $3N_{\text{nucl}} - 6$  normal coordinates (an exact parametrization).

No Born–Oppenheimer, no clamped nuclei, no neglect of mass polarization, no harmonic assumption.

### Outlook

(4.14) is the rigorous starting point: a Born–Huang expansion in the parametric electronic basis, then the Born–Oppenheimer truncation, recovers the familiar single-surface picture — a controlled approximation *of an exact operator*.

**The full molecular Hamiltonian:  
exact, quantized, and ready to approximate.**